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Explainable AI for critical care: a systematic review of interpretable models for sepsis and ICU mortality prediction

V. S. Athukorala¹ and W. M. K. S. Ilmini^{1*}

Purpose

Background Sepsis is a leading cause of mortality in intensive care units (ICUs), and its rapid progression poses significant challenges for early detection. Traditional scoring systems, such as SOFA and APACHE II, provide clinical benchmarks but often fail to capture subtle early signs of deterioration. Machine learning (ML) and deep learning (DL) models have demonstrated strong predictive performance; however, their “black-box” nature limits transparency, clinician trust, and adoption in real-world ICU settings. Explainable artificial intelligence (XAI) clarifies how predictions are derived.

Methodology This systematic review examines studies published between 2020 and 2025 that applied XAI methods, including SHAP, LIME, Grad-CAM, and sensitivity analysis, to predict sepsis onset and ICU mortality. We analyzed the datasets used (e.g., MIMIC-III/IV, Emory University Hospital, Ruijin Hospital), model architectures, interpretability strategies, and the clinical features most strongly associated with predictions.

Results Findings indicate that XAI-enhanced models not only maintain high predictive accuracy but also highlight clinically meaningful indicators such as respiratory rate, blood urea nitrogen (BUN), urine output, and the Glasgow Coma Scale (GCS), thereby improving clinician confidence and facilitating adoption.

Conclusion Despite these advances, challenges remain, including limited prospective evaluation, inconsistent interpretability metrics, and variable integration into clinical workflows. We conclude by recommending future research priorities, including real-world validation, user-centered design, and multimodal data integration, to ensure that XAI can reliably support timely and informed decision-making in critical care environments.

Keywords Sepsis, Intensive care unit (ICU), Explainable artificial intelligence (XAI), Deep learning, Predictive Modeling, Clinical decision support

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Introduction

Background

Artificial intelligence (AI) is increasingly reshaping healthcare by supporting clinicians in making informed, data-driven decisions and enabling more personalized patient care [1, 2]. As AI models become more sophisticated, understanding how predictions are made becomes more challenging. Explainable AI addresses this gap by providing interpretable insights into model behavior, linking algorithm outputs to clinical reasoning, and promoting trust in high-stakes clinical environments (Fig. 1).

While predictive accuracy is important, it is not sufficient on its own. Clinicians must also justify their decisions to patients, families, and regulatory bodies, highlighting the need for transparency in AI models. Techniques such as SHapley Additive exPlanations (SHAP) [4], Local Interpretable Model-agnostic Explanations (LIME) [5], and Gradient-weighted Class Activation Mapping (Grad-CAM) [6] provide clarity by revealing how models reach their conclusions, making AI outputs more actionable and practical in real ICU settings. Sepsis is a severe, life-threatening condition caused by a dysregulated immune response to infection and remains a major contributor to ICU mortality worldwide [7–9]. Although early recognition and timely intervention are critical for improving patient outcomes, commonly used clinical scoring systems such as the Sequential Organ Failure Assessment (SOFA) and the Acute Physiology and Chronic Health Evaluation II (APACHE II) were designed to quantify organ dysfunction and estimate mortality risk rather than to diagnose sepsis or support early detection. As a result, while these tools are valuable for assessing disease severity, they may be limited in identifying subtle early physiological deterioration within

complex ICU data [9]. Machine learning (ML) and deep learning (DL) models can leverage large ICU datasets, including Medical Information Mart for Intensive Care, Version IV (MIMIC-IV) [10], to detect patterns and predict sepsis onset more accurately than traditional scoring methods [11, 12]. However, these models often operate as “black boxes,” making it challenging for clinicians to trust the predictions and incorporate them into timely decision-making.

Figure 2 illustrates the difference between conventional scoring approaches and AI/XAI-based pipelines. Traditional scores rely on fixed rules and a limited set of clinical parameters, whereas AI/XAI pipelines utilize high-dimensional electronic health record data. By applying interpretability techniques, these models highlight the factors driving predictions, improving transparency and supporting safer, more personalized care in ICU settings.

The remainder of this paper is organized as follows. Section “[Methodology](#),” describes the design of the review, including the literature search strategy, databases and keywords used, inclusion and exclusion criteria, and the steps followed to analyze the selected studies. Section “[Literature Review](#),” summarizes recent research on XAI in sepsis and ICU mortality prediction, highlighting commonly used datasets, model types, and interpretability techniques. Section “[Results](#)” and “[Discussion](#),” presents the key findings from the reviewed studies, evaluates model performance and explainability, discusses clinical implications, and identifies existing research gaps and potential directions for future work. Finally, Section “[Discussion](#)” concludes the paper with a synthesis of insights and recommendations for advancing XAI in critical care applications.

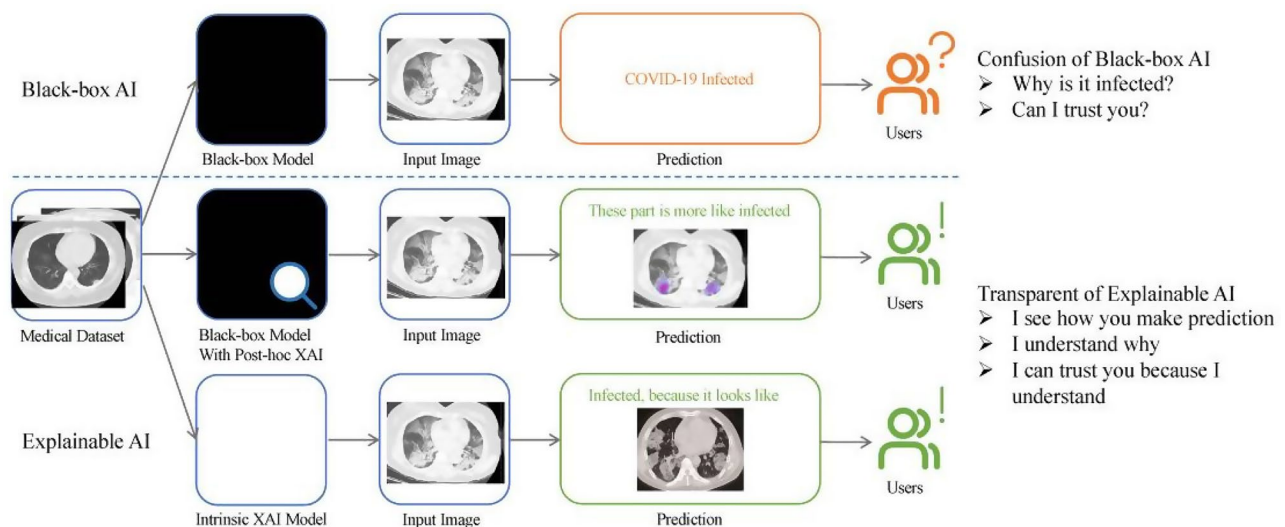


Fig. 1 Illustration of explainable AI (XAI) in ICU settings, showing how model transparency supports clinical decision-making (adapted from Chaddad et al. [3])

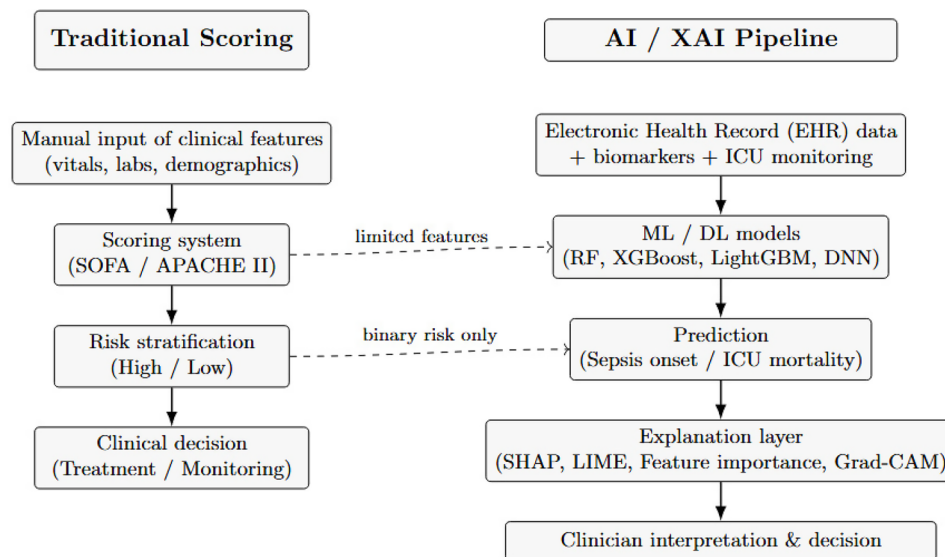


Fig. 2 Traditional scoring systems versus AI/XAI pipelines for sepsis and ICU mortality prediction

Problem statement

While machine learning (ML) and deep learning (DL) models have shown significant potential in predicting sepsis onset and patient outcomes [11–17], many of these systems function as “black boxes,” generating predictions without providing explanations for their reasoning. In critical care, where timely decisions can directly affect patient survival, this lack of transparency can reduce clinician confidence, delay necessary interventions, and limit the practical adoption of AI tools [1, 2].

Clinicians need predictive systems that are not only accurate but also interpretable. Explainable AI addresses this need by providing human-interpretable outputs and highlighting clinically important features that drive model decisions [4–6, 11–13]. Despite the promise of XAI, it remains unclear which techniques are most effective for predicting sepsis and ICU mortality in real-world clinical settings. Moreover, comprehensive comparisons of different XAI approaches across diverse ICU datasets are lacking [12, 13, 18]. Addressing these gaps is essential for developing AI tools that clinicians can trust and integrate reliably into their workflow.

Research gap

Although explainable artificial intelligence has demonstrated potential in supporting the prediction of sepsis and ICU mortality [4–6], its application in real clinical environments is still limited. Most published studies rely on retrospective data and are often based on a single institution or dataset [12–17]. This narrow scope restricts the generalizability of the findings and raises concerns about how these models would perform across diverse patient populations and healthcare settings. In addition, much of the current research prioritizes improving prediction

accuracy while giving less attention to whether the explanations produced by XAI models are truly understandable and useful for clinicians [1, 2, 12]. As a result, there is uncertainty about the practical value of these explanations in time-critical ICU situations. Another important limitation is the lack of clinician-centred evaluation. Only a small number of studies have examined how healthcare professionals interpret XAI outputs, how these explanations influence clinical decisions, or how easily such tools can be incorporated into existing ICU workflows. Furthermore, multi-centre validation and prospective testing remain rare [12, 13, 18], making it difficult to identify which interpretability techniques are most reliable in real-world practice. These gaps underline the need for future research that goes beyond model performance alone. There is a clear requirement for studies that integrate external validation, clinician involvement, and rigorous assessment of interpretability. Developing XAI models that are both accurate and genuinely understandable in varied ICU environments is essential for safe and effective clinical adoption, ultimately supporting better decision-making and improved patient outcomes.

Aim and objectives

This study aims to investigate how explainable artificial intelligence can enhance both the predictive accuracy and interpretability of models used for sepsis onset and ICU mortality. By leveraging large-scale and diverse datasets, including MIMIC-IV, Emory University Hospital, UCSD, and Ruijin Hospital [10], the study evaluates various machine learning and deep learning approaches to uncover clinically meaningful patterns in patient data [11–17]. Explainability techniques, such as SHAP, LIME, Grad-CAM, feature importance, and sensitivity analysis

[4–6], are applied to reveal the key factors influencing predictions and to align model outputs with established clinical scoring systems like SOFA and APACHE II [7, 9]. The study also examines model performance using relevant metrics, including accuracy and the area under the receiver operating characteristic curve Area Under the Curve (AUC), and identifies limitations that may impede the practical adoption of XAI in ICU settings. Finally, emerging methods, including argumentation-based explainability [19, 20], are explored as promising strategies to improve transparency, foster clinician trust, and support effective integration of AI tools into critical care workflows.

Methodology

This systematic review examines the role of explainable and interpretable machine learning approaches in supporting clinical decision-making within adult intensive care units (ICUs), with particular attention to predicting sepsis onset, ICU readmissions, and in-hospital mortality. The review is conducted following a rigorous and reproducible protocol, designed to ensure methodological transparency, reduce potential bias, and facilitate replication of the findings.

Review protocol

This review was conducted following the PRISMA 2020 guidelines, employing a structured, stepwise methodology to ensure methodological transparency, reproducibility, and rigor.

Step 1: Define Research Questions The initial stage of this review focused on formulating the research questions that would direct the systematic search and subsequent analysis. The primary objective was to investigate how explainable artificial intelligence and interpretable machine learning methods are utilized for sepsis prediction in adult intensive care units. Additionally, the review aimed to identify the clinical tasks addressed by XAI, including sepsis onset prediction, ICU readmissions, and in-hospital mortality. Finally, it examined which datasets, types of predictive models, and interpretability techniques are most frequently applied in these studies.

Step 2: Literature Search A systematic search was conducted across several electronic databases to identify studies published between January 2020 and January 2025. The primary sources included ResearchGate, IEEE Xplore, Scopus, Web of Science, and Google Scholar. To ensure completeness, an additional search using Google Scholar was performed to capture studies that may have been missed in the primary search. This comprehensive approach allowed for the inclusion of a broad and representative set of studies relevant to the topic.

The search strategy was guided by a set of predefined keywords, applied individually and in combination to

maximize coverage. Keywords included “Artificial intelligence in ICU,” “Sepsis prediction using machine learning,” “ICU readmission model,” “Clinical decision support systems,” “Explainable artificial intelligence,” “Interpretable machine learning in ICU,” and “Mortality prediction.” This method facilitated the identification of studies focused on AI applications, interpretable models, and predictive analytics in critical care settings.

Step 3: Study Selection Screening was conducted in two phases, independently by two reviewers to minimize bias:

1. *Title and Abstract Screening:* In this stage, studies with irrelevant topics or lacking interpretability components were excluded. Duplicate records had already been removed during the identification phase, ensuring that only unique studies were considered for further review.
2. *Full-Text Review:* Assess remaining studies against predefined inclusion and exclusion criteria. Disagreements were resolved through discussion, with a third senior reviewer adjudicating when necessary.

Inclusion Criteria:

- Sepsis, severe sepsis, or septic shock, defined according to Sepsis-3 criteria or equivalent clinical definitions
- Use of interpretable/explainable AI/ML techniques (e.g., SHAP, LIME, GradCAM, feature importance)
- Prediction of sepsis onset, ICU readmission, or in-hospital mortality with defined outcome horizons
- Use of ICU or Electronic Health Record (EHR) datasets (e.g., MIMIC-III, MIMIC-IV, Emory University Hospital, University of California, San Diego (UCSD), RH)
- Peer-reviewed articles, conference papers, English language, published 2020–2025
- Filters applied: Peer-reviewed journal and conference publications reporting original research relevant to the study objectives.

Exclusion Criteria:

- Pediatric or neonatal ICUs
- Studies without clear sepsis definition or outcome horizon
- Black-box-only models without interpretability
- Non-ICU or unrelated clinical tasks
- Abstracts without full text or inaccessible articles

Step 4: Data Extraction Data were independently extracted by two reviewers using a structured form, capturing:

- Author names and publication year
- Dataset used
- Clinical task addressed
- Type of ML or DL model
- Explainability technique applied
- Key features identified as influential

The detailed study selection process is illustrated in the PRISMA 2020 flow diagram in Fig. 3, showing the number of records identified, screened, excluded, and included at each stage.

Our initial database search identified 100 potentially relevant records. After removing 10 duplicate entries, 90 records remained for screening based on titles and abstracts. During this screening stage, 47 records were excluded for clearly defined reasons, including lack of relevance to the study topic, failure to meet the required study design criteria, or insufficient data for meaningful analysis. The remaining 43 full-text articles were then assessed for eligibility, and all 43 studies satisfied the inclusion criteria and were therefore included in the qualitative synthesis (Fig. 3). This structured selection process ensures transparency and enables readers to clearly follow how the final set of studies was identified.

Literature review

In recent years, explainable artificial intelligence has gained attention as a way to address the limitations of “black-box” models in healthcare. This is especially important in conditions like sepsis, where decisions must be made quickly and clinicians need to understand why an algorithm is making a particular prediction. Sepsis and ICU mortality remain among the most complex and time-sensitive challenges in medicine, and researchers

have increasingly explored how explainability can make AI systems more trustworthy, transparent, and clinically useful.

A comprehensive review by Nicolaou et al. [21] outlined how XAI methods such as SHAP [22], LIME [5], and Grad-CAM [6] are being applied to sepsis prediction. These tools do more than provide predictions—they allow clinicians to see which variables drive those predictions, making AI outputs easier to trust in high-stakes environments.

Other reviews have expanded this perspective. For instance, Moazemi et al. [23] focused on cardiovascular ICUs and demonstrated that AI-powered decision support systems, when combined with explainability, can help physicians interpret results with greater confidence. However, they also pointed out a recurring issue: while models often work well in research, their integration into everyday hospital workflows remains limited.

Several studies have taken a more technical approach. Alabdulhafith et al. [19] developed a hybrid ICU readmission model that combined ensemble learning with particle swarm optimization. By embedding XAI techniques, the model not only delivered strong predictions but also highlighted clinical markers like blood pressure and heart rate—information that clinicians can directly verify against patient records.

Jiang et al. [24] explored mortality risk among patients who survived sepsis but required ICU readmission. Their explainable model revealed key patterns in patient data, such as low oxygen saturation and high creatinine levels, which are well-known clinical risk factors. The incorporation of explainable AI (XAI) has enhanced both the accuracy and clinical relevance of predictive models. Transferability is another key consideration in AI

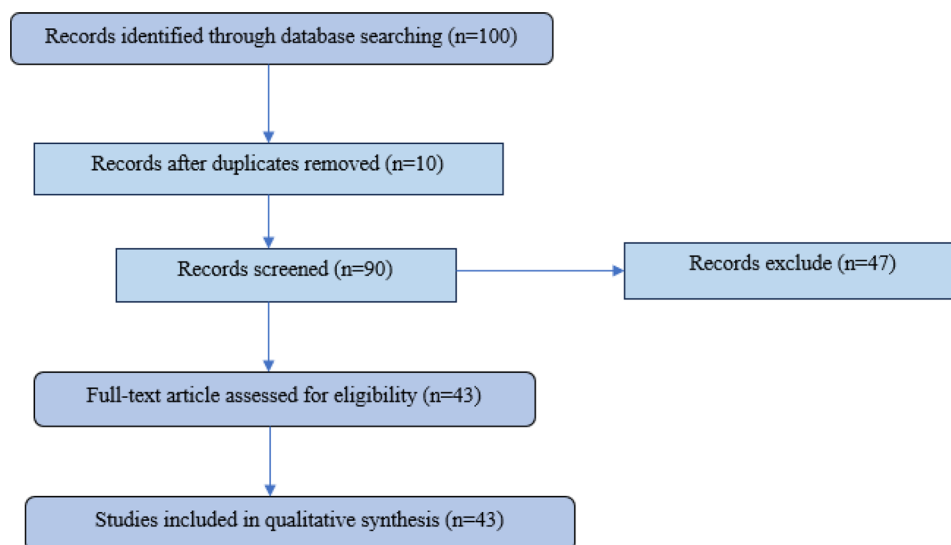


Fig. 3 PRISMA 2020 flow diagram of study selection

applications. For instance, Chen et al. [18] evaluated sepsis prediction models across multiple hospitals and found that interpretable models consistently outperformed opaque systems when applied in new clinical settings. These results indicate that transparency not only fosters clinician trust but also improves adaptability across diverse healthcare environments.

Beyond sepsis prediction, Lu et al. [20] applied interpretable machine learning to investigate sepsis-associated encephalopathy, a frequently overlooked complication. Their approach identified early physiological and cognitive markers, providing clinicians with actionable predictive insights and opportunities for timely intervention.

Lightweight models have also demonstrated significant utility. Nesaragi and Patidar [14] showed that even basic ICU measures, such as heart rate and body temperature, when combined with explainable algorithms, can facilitate early detection of sepsis. This underscores the potential of XAI in resource-limited settings where complex datasets may be unavailable.

Several studies, including those by Hu et al. [13, 16], Li et al. [15], and Ke et al. [17], highlight the necessity of validating XAI models using real-world ICU data. Their findings demonstrate that integrating clinical expertise with model interpretability produces systems that are both accurate and practical for frontline healthcare providers. Various approaches to model explainability continue to be explored to further enhance clinical decision support. Chen and Hernández [25], for example, applied sensitivity analysis to understand how small changes in input variables influenced sepsis predictions. This technique provided reassurance that the models were learning from clinically relevant patterns rather than spurious associations.

Across the literature, datasets such as MIMIC-IV [10] and hospital-specific records (e.g., Emory, UCSD, and Ruijin hospitals) have been central to developing and validating these models. Among the most widely used interpretability methods are SHAP [4], LIME [5], sensitivity analysis [25], and visualization techniques like Grad-CAM [6].

While these developments are promising, challenges remain. Most studies are retrospective, with few tested in real-time hospital settings. Many also prioritize performance metrics such as AUC over usability and interpretability. Moving forward, researchers must focus on external validation, multi-center prospective studies, and clinician-centered evaluations to ensure that XAI models can be safely and effectively integrated into ICU decision-making.

Recent research highlights the growing role of explainable artificial intelligence in healthcare, especially for predicting sepsis. According to Sadeghi et al. [26], XAI helps clinicians see how complex AI models make decisions.

They point out that even highly accurate models are of limited use in ICU settings if medical staff cannot understand or trust the reasoning behind the predictions.

Explainable AI is becoming increasingly important in healthcare, especially for predicting sepsis. Sadeghi et al. [27] note that XAI helps clinicians understand how AI models make decisions. They emphasize that even accurate models are of little use if doctors cannot trust or interpret the predictions. By making AI transparent, XAI supports better decisions and encourages real-world adoption in hospitals.

Recent studies have highlighted the increasing potential of machine learning and deep learning techniques for predicting mortality in patients with sepsis. Li and Liu [28] developed a deep learning model using ICU patient data to identify individuals at high risk of in-hospital death. They emphasized that predictive accuracy alone is not sufficient if clinicians cannot interpret the results. By integrating explainable AI methods, their model identifies the most influential patient factors, supporting both clinical understanding and trust in the model's outputs.

Similarly, Zhang et al. [29] reviewed multiple machine learning approaches for sepsis mortality prediction. Their findings showed that although many models achieve high accuracy, performance varies depending on data quality and feature selection. They stressed that interpretability is essential for clinical use, as clinicians must understand which variables drive predictions in order to make informed decisions in critical care settings.

Beyond mortality prediction, machine learning has also been widely applied to detect sepsis and identify high-risk ICU patients. Fleuren et al. [30] demonstrated that while predictive models often perform well, transparency regarding the contributing patient factors is critical for clinical trust. Wang et al. [31] similarly developed a predictive framework that achieved strong accuracy while clearly highlighting key clinical features influencing outcomes, reinforcing the importance of interpretable results for effective medical decision-making.

Systematic evaluations further support these observations. Yang et al. [32] conducted a comprehensive review and meta-analysis of machine learning approaches for early sepsis detection and found that model reliability strongly depends on data quality and feature engineering. They concluded that interpretability should be a fundamental design requirement, as understandable models allow clinicians to confidently integrate AI tools into routine ICU practice.

Recent research has increasingly emphasized explainable and interpretable machine learning for mortality prediction. Gao et al. [33] developed models that estimated death risk using detailed clinical variables while highlighting the most influential predictors. Likewise, Wang et al. [34] proposed an interpretable framework

that visually represented contributing clinical factors, demonstrating how transparency can support timely and well-informed clinical decisions.

The value of interpretability has also been demonstrated across diverse healthcare settings. Diwan et al. [35] applied explainable machine learning in low- and middle-income countries, showing that transparent models can be effectively used even in resource-constrained ICUs. Their results highlight that interpretability not only builds clinician trust but also facilitates the practical implementation of AI systems in real-world clinical workflows.

Emergency and triage settings also benefit from interpretable machine learning for improving early sepsis prediction. Liu et al. [36] developed a model capable of identifying high-risk patients at early stages while providing clear explanations of the clinical factors driving each prediction. Similarly, Zilker et al. [37] introduced a decision-support framework for predicting ICU admission in patients with suspected sepsis, combining robust predictive performance with transparent feature attribution to support clinical planning and patient management.

Furthermore, Stylianides et al. [38] present significant advancements in AI applications within intensive care units, emphasizing sepsis prediction. Their study illustrates how explainable machine learning models can provide clear and interpretable insights into patient risk, allowing clinicians to make timely and informed decisions. By combining real-time patient monitoring with transparent AI outputs, these approaches improve the quality of care and support proactive intervention strategies.

Figure 4 presents a clear and integrated summary of the interpretable machine learning models reported in existing sepsis prediction studies. It brings together the main clinical prediction tasks explored in the literature such as early identification of sepsis, ICU mortality prediction, and length of stay estimation along with the corresponding modeling techniques and referenced works. By linking different model categories to their specific clinical objectives, the figure helps reveal common research patterns and shows how interpretability has been incorporated across diverse predictive tasks. In doing so, it complements the preceding literature review by providing a structured visual synthesis of prior research and

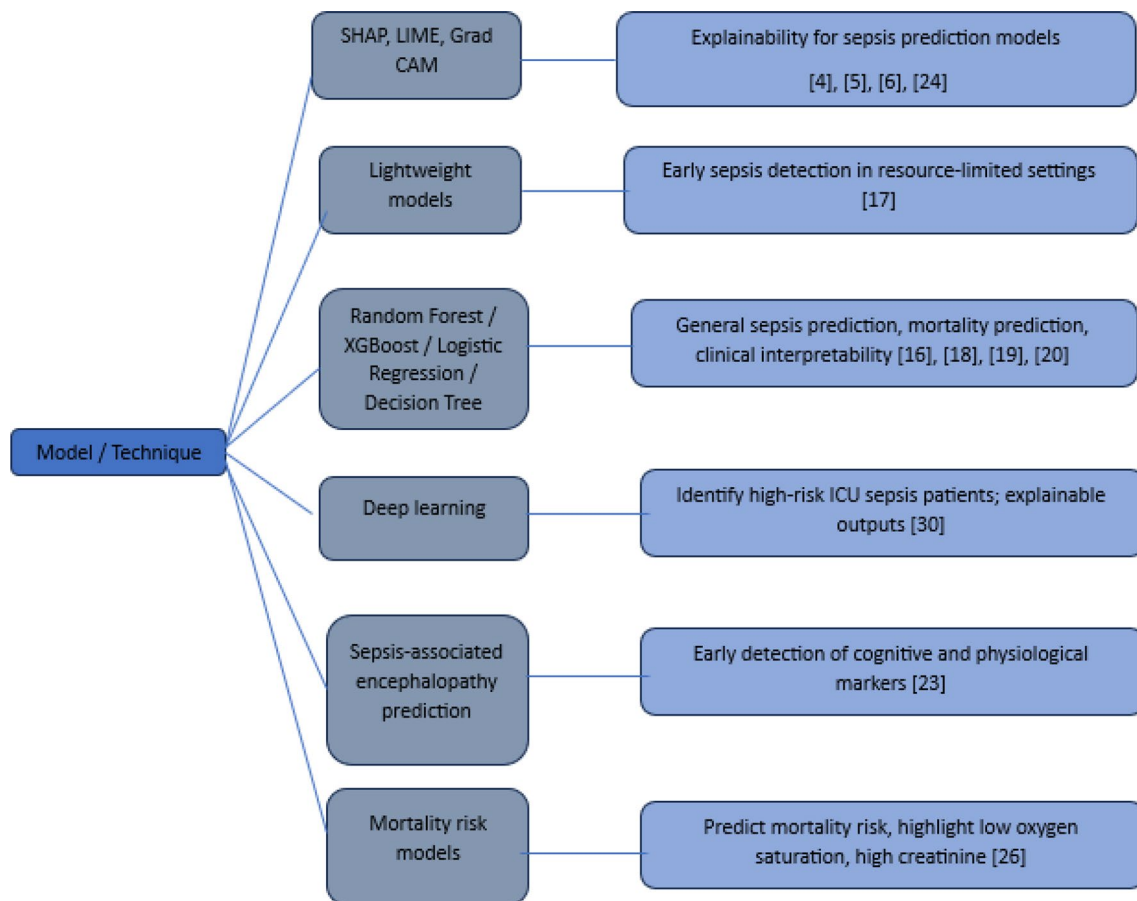


Fig. 4 Overview of interpretable machine learning models for sepsis prediction. The diagram shows the models, their predictive tasks (e.g., ICU mortality, early sepsis onset), and the associated references

Table 1 Summary of selected studies on explainable AI in sepsis and ICU prediction tasks, including dataset, cohort size, prediction horizon, model type, validation, metrics, XAI methods, and top features

Ref	Clinical Task	Dataset	Co-hort Size	Prediction Horizon	Model Type	Validation Type	Metrics	XAI Method	Explainability Metric	Top Features Identified
[4]	Mortality in sepsis survivors	MIMIC-III	12,300	30 days	Interpretable ML	Temporal Validation	AUC 0.87, Calibration 0.92	SHAP, Clustering	Stability	Creatinine, SOFA, O ₂ saturation
[7]	ICU readmission prediction	Custom ICU	5,000	7 days	Ensemble ML	Temporal Validation	AUC 0.86	Feature Importance	None	Vitals, GCS, Lab tests
[12]	Early sepsis prediction	MIMIC-IV	14,500	6 h	XGBoost	Train–Test Split	AUC 0.89	SHAP	None	MAP, Platelets, Urine Output
[13]	Sepsis prognosis	ICU multi-center	12,000	24 h	Random Forest	5-fold CV	AUC 0.87	SHAP	Stability	HR, Temp, WBC
[14]	Early sepsis detection	ICU dataset	8,500	6 h	Rule-based ML	Train–Test Split	AUC 0.86	LIME	Fidelity	HR, Temp, WBC
[15]	In-hospital mortality	ICU data	9,200	24 h	Interpretable ML	Temporal Validation	AUC 0.88	SHAP	Stability	SOFA, Creatinine, RR
[16]	ICU readmission mortality	Multi-center ICU	7,800	48 h	XGBoost	Train–Test Split	AUC 0.87	SHAP	None	MAP, HR, Lactate
[17]	Mortality prediction	MIMIC-IV	9,800	24 h	Random Forest	5-fold CV	AUC 0.84, AUCPR 0.78	SHAP, LIME	Fidelity	MAP, Platelets, Urine Output
[18]	Model transferability	ICU multi-center	10,000	12 h	Gradient Boosting	Temporal Split	AUC 0.85	SHAP	None	Lactate, MAP, Creatinine
[19]	Sepsis onset & mortality	Multi-center ICU	10,500	24 h	Ensemble ML + PSO	5-fold CV	AUC 0.88, AUCPR 0.82 ¹	SHAP, Grad-CAM	Fidelity, Stability	RR, GCS, BUN, Temp
[20]	Sepsis-associated encephalopathy	ICU data	8,200	48 h	Interpretable ML	Train–Test Split	AUC 0.85, good calibration	SHAP	Fidelity	Lactate, GCS, MAP
[23]	Cardiovascular ICU monitoring	Multi-center	4,500	24 h	Ensemble ML	5-fold CV	AUC 0.86	SHAP	Fidelity	HR, MAP, Lactate
[24]	Sepsis survivor risk	MIMIC-III	8,100	30 d	Interpretable ML	Temporal Validation	AUC 0.86	SHAP, Clustering	Stability	Creatinine, SOFA, Oxygen
[25]	Sepsis detection	ICU data	6,700	12 h	DNN	Train–Test Split	AUC 0.85	Sensitivity Analysis	None	HR, Temp, WBC
[39]	Real-time sepsis prediction	ICU dataset	5,500	6 h	Gradient Boosting	Train–Test Split	AUC 0.85	SHAP	Fidelity	HR, Temp, WBC

Abbreviations: RR = Respiratory Rate; GCS = Glasgow Coma Scale; BUN = Blood Urea Nitrogen; Temp = Temperature; MAP = Mean Arterial Pressure; WBC = White Blood Cells; SOFA = Sequential Organ Failure Assessment; HR = Heart Rate; Lactate = Blood Lactate Level; Creatinine = Serum Creatinine; Platelets = Platelet Count; Urine Output = Urine Output Volume; Vitals = Vital Signs (HR, BP, Temp, RR); Lab tests = Laboratory Tests

¹AUCPR: Area under the Precision-Recall curve

by highlighting areas where certain prediction tasks or interpretable models remain insufficiently investigated.

Results

Table 1 contains a summary of selected studies on explainable AI in sepsis and ICU prediction tasks, including dataset, cohort size, prediction horizon, model type, validation, metrics, XAI methods, and top features.

Summary of included studies

A total of 43 high-quality studies published between 2020 and 2025 were included in this review. These studies focused on predicting sepsis onset, ICU mortality, and readmission risk using interpretable machine learning

models. Table 1 summarizes 15 representative studies selected from the total of 43 included in this review, highlighting key methodologies and findings for clarity and conciseness. Most studies used real-world datasets such as MIMIC-III/IV, Emory University Hospital (EUH), and Ruijin Hospital EHRs. Explainability methods like SHAP, LIME, GradCAM, Feature Importance, and Sensitivity Analysis were commonly applied. The models used included random forests, XGBoost, deep neural networks, and ensemble techniques.

Risk of bias result

The methodological quality of the included studies was systematically evaluated using the PROBAST tool,

which assesses potential sources of bias across four key domains: Participants, Predictors, Outcomes, and Analysis. Each study was independently reviewed by two researchers to ensure objectivity and consistency, and any discrepancies were resolved through discussion until consensus was reached.

In the Participants domain, most studies (e.g. [3, 7, 11, 12]), demonstrated a low risk of bias, indicating that the study populations were carefully selected and accurately represented real-world ICU patients. Clear inclusion and exclusion criteria helped minimize selection bias and improve generalizability.

For the Predictors domain, several studies (e.g. [13, 15, 16, 18]), exhibited an unclear or moderate risk of bias. This was primarily due to inconsistent reporting of predictor measurement methods, insufficient detail regarding predictor selection, or lack of standardization across patients.

In the Outcomes domain, most studies (e.g. [14, 17, 20]), maintained a low risk of bias, with clearly defined and consistently measured outcomes such as sepsis onset, ICU mortality, or readmission events. Independent and objective outcome assessment further strengthened the reliability of results.

The Analysis domain revealed more variability in methodological rigor. While some studies applied robust statistical validation, appropriate handling of missing data, and measures to prevent overfitting (e.g. [34–36], others presented unclear or high risk of bias due to inadequate model validation, lack of external validation, or insufficient reporting of analytical methods (e.g. [25, 38]).

Overall, this assessment highlights that while many included studies employed rigorous methods, there remains a need for improved transparency and consistency, particularly in predictor measurement and analytical strategies. These findings provide important context when interpreting the reported performance and applicability of machine learning models for sepsis prediction.

Evidence synthesis

This section summarizes how explainable artificial intelligence (XAI) models perform in predicting sepsis onset and ICU mortality. It focuses on the types of validation used, reported performance metrics, key predictive features, and the interpretability methods applied across the included studies.

Predictive performance by validation type

- Studies using internal validation approaches, such as k-fold cross-validation or random train–test splits within a single dataset, generally demonstrated strong predictive performance. For instance, Diwan et al. developed explainable machine learning

models for predicting sepsis mortality in a tertiary care hospital ICU, where the Extra Trees classifier achieved an AUROC of 0.87 and maintained consistent results across the cross-validation folds. [35].

- In contrast, studies that applied external or generalizable validation strategies typically showed slightly lower predictive performance but provided stronger evidence of model applicability across different settings. For example, Li et al. developed and validated an explainable machine learning model using both internal and external datasets, achieving an AUROC of 0.87 on internal validation and 0.81 on external validation, highlighting how model performance can differ when tested on independent cohorts. [39].
- Other research using large critical care datasets has similarly demonstrated strong predictive performance across various validation approaches. For example, Zhang et al. applied multiple machine learning models to the MIMIC-IV dataset, with the Random Forest classifier achieving an AUROC of up to 0.94 in internal evaluations, emphasizing that both the choice of algorithm and the selection of predictive features play a key role in determining model effectiveness [36].

Top predictive features

Across studies, the most frequently identified predictive features included:

- Respiratory Rate (RR)
- Glasgow Coma Scale (GCS)
- Blood Urea Nitrogen (BUN)
- Urine Output
- Mean Arterial Pressure (MAP)

These features were consistently highlighted as significant predictors of sepsis onset and ICU mortality.

Interpretability methods

The most commonly applied interpretability methods were:

- SHAP (Shapley Additive Explanations)
- LIME (Local Interpretable Model-agnostic Explanations)
- Sensitivity Analysis

These methods were utilized to enhance the transparency and trustworthiness of machine learning models in clinical settings.

Discussion

XAI-enhanced models have consistently shown strong predictive performance in ICU settings, often achieving AUC (Area Under the Receiver Operating Characteristic Curve) values above 0.85 [7, 12, 19]. The AUC measures how well a model can distinguish between high-risk patients, such as those who are likely to develop sepsis or experience mortality, and lower-risk patients, with higher values indicating more reliable predictions. Among interpretability methods, SHAP is the most widely used, helping to identify key clinical characteristics such as respiratory rate, blood urea nitrogen, and Glasgow Coma Scale (GCS) that drive model decisions [19, 22]. Models trained on large data sets such as MIMIC-III and MIMIC-IV have demonstrated strong generalizability, suggesting that their predictions can be applied to diverse ICU populations [12, 14].

The studies reviewed demonstrate that explainable AI (XAI) models consistently achieve high predictive performance across a range of ICU contexts and sepsis-related clinical tasks. Most models reported AUC values exceeding 0.84, with some attaining up to 0.89 for early sepsis prediction [7, 12, 13, 17, 19, 22]. For example, an XGBoost model trained on the MIMIC-IV dataset for early sepsis prediction achieved an AUC of 0.89 [12], whereas an ensemble machine learning model predicting ICU readmissions reached an AUC of 0.86 [7]. Similarly, models developed for sepsis onset, in-hospital mortality, and cardiovascular ICU monitoring exhibited high predictive accuracy, highlighting the robustness and reliability of XAI approaches in critical care settings [19, 22, 23].

Interpretability methods were widely applied across these studies, with SHAP emerging as the most frequently used technique [12, 13, 15, 16, 18, 19, 22, 24, 39]. Complementary approaches, including LIME and Grad-CAM, were implemented in select studies to provide additional insights into model behavior [14, 17, 19], while sensitivity analysis was specifically employed in sepsis detection with deep neural networks [25]. These interpretability strategies enabled the identification of key physiological variables driving model predictions, such as respiratory rate (RR), Glasgow Coma Scale, mean arterial pressure (MAP), lactate, creatinine, heart rate (HR), blood urea nitrogen (BUN), body temperature, platelet count, and urine output [7, 12–18, 22, 24, 25, 39]. By highlighting these clinically relevant factors, XAI methods provide transparent explanations that align with clinicians' reasoning and support informed decision-making in ICU settings.

Despite these encouraging results, several important limitations remain. Most studies relied on retrospective datasets, including MIMIC-III, MIMIC-IV, and other ICU databases, with cohort sizes ranging from 4,500 to 14,500 patients ([12, 13, 18, 19, 22, 23]). Only a limited

number of studies conducted external or temporal validation ([7, 15, 18, 22, 24]), and very few models have been prospectively implemented in real clinical environments. Furthermore, while XAI methods offer valuable interpretability, the lack of standardized metrics for assessing explainability across studies complicates the comparison of different approaches and limits the evaluation of their reliability in real-world ICU settings ([2, 20]).

In conclusion, XAI-enhanced models offer significant potential to boost clinician confidence and facilitate the integration of AI-driven decision support in ICU settings. Nevertheless, it is crucial to be aware of the limitations inherent to these methods. XAI techniques may unintentionally capture noise from overfitted models or artifacts in the data, producing explanations that highlight irrelevant patterns rather than truly meaningful clinical signals. As noted by Spitzer et al. [40], such misleading explanations can create a “misinformation effect,” where users might draw incorrect conclusions about the reasoning behind AI predictions. For this reason, while XAI contributes to transparency, clinicians should interpret its outputs carefully, always combining them with thorough patient assessment and professional judgment to guide clinical decision-making effectively. By highlighting physiologically meaningful risk factors, these models provide interpretable insights that complement, rather than replace, clinical judgment. This allows healthcare professionals to act on timely, data-driven information, ultimately enhancing patient outcomes. The studies summarized in the table indicate that combining ensemble and interpretable machine learning approaches with SHAP or similar XAI methods can achieve both high predictive accuracy and transparent, actionable explanations, reinforcing their utility for effective and trustworthy critical care management.

Conclusion

This systematic review explored the use of explainable artificial intelligence for predicting sepsis onset and ICU mortality, with particular emphasis on improving both predictive accuracy and interpretability in critical care settings. The review shows that much of the current research has been conducted in high-income countries; however, recent studies indicate that XAI-based approaches can also be adapted to resource-constrained healthcare environments. This suggests a broader potential for clinical adoption, while also highlighting practical challenges related to real-world implementation.

Across the reviewed literature, XAI techniques—most commonly feature attribution methods—were widely used to improve model transparency and to link predictions with clinically relevant indicators, including vital signs, neurological status, and renal function measures. Overall, these models demonstrated strong predictive

performance while providing interpretable outputs that may assist clinicians in understanding and acting upon model predictions in ICU settings. Common methodological trends were also observed, such as the frequent combination of ensemble or deep learning models with post hoc explainability techniques to achieve a balance between accuracy and transparency.

Despite these encouraging findings, several limitations remain. The majority of studies relied on retrospective, single-center datasets, and there was considerable variability in how interpretability was evaluated, limiting meaningful comparisons across models. In addition, evidence from prospective studies, real-world deployments, and clinician-centered evaluations of XAI systems remains limited.

In summary, this review underscores the growing importance of XAI in supporting critical care prediction tasks while emphasizing the need for future work focused on external validation, standardized approaches to interpretability assessment, and effective integration into clinical workflows. Addressing these challenges will be crucial to ensure that explainable AI systems can be safely, effectively, and meaningfully adopted in routine ICU practice.

Future work

Future research should aim to make explainable AI a practical and trusted tool in hospitals. Standardized metrics for fidelity and stability can help assess model reliability [8, 21] while prospective ICU trials are needed to evaluate real-world performance and impact on clinical decisions [13, 23]. Engaging clinicians in design and integrating multimodal data such as lab results, imaging, and clinical notes can improve both the accuracy and clarity of predictions [2, 13]. Advanced reasoning approaches, like argumentation-based AI, may further allow models to explain not just what they predict, but why, fostering trust and supporting better patient care [14, 23].

In addition, future investigations should incorporate the principles of responsible AI, with particular attention to fairness and the mitigation of bias. Many commonly used ICU datasets, such as MIMIC-III and MIMIC-IV, are drawn from relatively narrow patient populations and may not adequately reflect the diversity found across different regions, ethnic groups, and healthcare systems. To promote broader applicability, researchers should prioritize the use of more representative datasets, apply fairness-aware modeling strategies, and adopt transparent reporting standards. Equally important is close collaboration with clinicians, whose expertise is essential for validating model outputs and ensuring that interpretability methods evolve into reliable tools that can be confidently applied in everyday clinical practice.

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Author contributions

V. S. Athukorala conducted the survey, wrote, and edited the manuscript. W. M. K. S. Ilmini supervised the work and contributed to writing and editing the manuscript.

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Ethics approval and consent to participate

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Competing interests

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